Janvijay Singh

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Research Focus

My research focuses on reliable reasoning and evaluation in LLMs and LLM agents, including how their capabilities and behaviors are shaped by post-training. My work spans LLM reasoning [C7,P1], automated evaluation [C5,C6], and post-training induced behaviors [C8,C9], with an emphasis on faithfulness and generalization.

Education

University of Illinois Urbana-Champaign — Urbana, IL
PhD in Computer Science | Advisors: Dilek Hakkani-Tür and Gokhan Tur — *Aug 2024 – Dec 2027*

Georgia Institute of Technology — Atlanta, GA
MS in Computer Science | Machine Learning Specialisation; GPA: 3.91/4.00 — *Aug 2021 – May 2023*

Indian Institute of Technology — Varanasi, India
BTech in Computer Science and Engineering | GPA: 9.62/10.00; Department rank: 2/65 — *Jul 2014 – May 2018*

Research Experience

Microsoft Research — May 2026 – Oct 2026
Research Intern, AI Frontiers | Mentors: Asli Çelikyilmaz and Ece Kamar — *Redmond, WA*

Adaptation in Terminal Agents. Built a scalable pipeline to generate novel environments for studying adaptation; investigated whether post-training with novelty augmentation and meta-action reasoning improves adaptation [P1].

Salesforce Research — May 2025 – Aug 2025
AI Research Intern, LLM Reasoning | Mentors: Austin Xu and Shafiq Joty — *Palo Alto, CA*

Reliability of LLM Judges. Studied whether fine-tuned LLM judges stay reliable on responses from weaker and stronger generators and on unseen questions, across SFT and DPO based training [C6]; contributed to analyzing how judge accuracy varies with problem difficulty, generator strength, and judge strength [C5].

University of Illinois Urbana-Champaign — Aug 2024 – Present
Graduate Researcher | Advisors: Dilek Hakkani-Tür and Gokhan Tur — *Urbana, IL*

Functional Importance of Reasoning. Showed via greedy pruning and attention analysis that LLMs encode functional importance of reasoning tokens; this importance supports effective compressed-reasoning distillation [C7].

Post-Training-Induced LLM Behaviors. Identified alignment-induced unfaithfulness: LLMs silently trade task-faithfulness for alignment and this worsens with model size and chain-of-thought reasoning [C8]; localized and ablated the residual-stream direction for "AI-like" writing style, lowering AI-detection rates for several LLMs [C9].

Evaluation of Interactive Agents. Developed a multi-stage evaluation framework that uses atomic LLM-judged criteria to estimate user satisfaction with interactive planning agents [J1].

Georgia Institute of Technology — Aug 2021 – May 2023
Graduate Researcher | Advisors: Fan Bai, Diyi Yang, and Saif M. Mohammad — *Atlanta, GA*

Citation Analysis of the NLP Community. Analyzed ∼70K NLP papers, revealing low temporal citation diversity [C3] and high geographic disparity in paper acceptance and citations [C1].

Entity Tracking in Procedural Text. Reformulated entity-state tracking as a QA task and trained LMs with mention-guided decoding, outperforming existing task-specific approaches [C2].

ETH Zürich — Jul 2022 – Sep 2022
Summer Research Fellow, AI in Education | Advisor: Mrinmaya Sachan — *Zürich, Switzerland*

Visual Enrichment of Textbooks. Framed adding web images to textbooks as a text–image matching problem; evaluated the effectiveness of VLMs in image assignment based on relevance and diversity [C4].

Selected Publications

[=] *denotes equal contribution. C: conference, J: journal, P: preprint. Paper titles are linked.*

[P1] **Janvijay Singh**, Vaish Shrivastava, Dilek Hakkani-Tür, Ece Kamar, Asli Çelikyilmaz. *When Successful Strategies Fail: Adaptation to Environmental Novelty in Terminal Agents*. **Under review at ICLR 2027**.

[C9] Aniket Anand[=], **Janvijay Singh**[=], Zhewei Sun, Dilek Hakkani-Tür, Nick Feamster. *Measuring, Localizing, and Ablating Alignment Signatures in LLMs*. **EMNLP**, 2026.

[C8] Pardis Sadat Zahraei, **Janvijay Singh**, Gokhan Tur, Dilek Hakkani-Tür. *Emergent Unfaithfulness: How Alignment Training Causes Language Models to Silently Override Task Faithfulness*. **COLM; KnowFM at ACL**, 2026. **Outstanding Paper Award (KnowFM).**

[C7] **Janvijay Singh**, Dilek Hakkani-Tür. *Do LLMs Encode Functional Importance of Reasoning Tokens?* **ACL**, 2026.

[C6] **Janvijay Singh**, Austin Xu, Yilun Zhou, Yefan Zhou, Dilek Hakkani-Tür, Shafiq Joty. *On the Shelf Life of Finetuned LLM-Judges: Future Proofing, Backward Compatibility, and Question Generalization*. **ICLR**, 2026.

[C5] Yefan Zhou, Austin Xu, Yilun Zhou, **Janvijay Singh**, Jiang Gui, Shafiq Joty. *Variation in Verification: Understanding Verification Dynamics in Large Language Models*. **ICLR**, 2026.

[J1] Takyoung Kim[=], **Janvijay Singh**[=], Shuhaib Mehri[=], Emre Can Acikgoz, Sagnik Mukherjee, Nimet Beyza Bozdag, Sumuk Shashidhar, Gokhan Tur, Dilek Hakkani-Tür. *AURA: A Diagnostic Framework for Tracking User Satisfaction of Interactive Planning Agents*. **IEEE/ACM TASLP**, 2026; **MTI-LLM at NeurIPS**, 2025.

[C4] **Janvijay Singh**, Vilém Zouhar, Mrinmaya Sachan. *Enhancing Textbooks with Visuals from the Web for Improved Learning*. **EMNLP**, 2023.

[C3] **Janvijay Singh**[=], Mukund Rungta[=], Diyi Yang, Saif M. Mohammad. *Forgotten Knowledge: Examining the Citational Amnesia in NLP*. **ACL**, 2023. **Best Paper Honorable Mention.**

[C2] **Janvijay Singh**, Fan Bai, Zhen Wang. *Entity Tracking via Effective Use of Multi-Task Learning Model and Mention-Guided Decoding*. **EACL**, 2023.

[C1] Mukund Rungta[=], **Janvijay Singh**[=], Saif M. Mohammad, Diyi Yang. *Geographic Citation Gaps in NLP Research*. **EMNLP**, 2022.

Additional Experience

TikTok Jan 2024 – Jul 2024

Machine Learning Engineer 2, Social Recommendation *San Jose, CA*

Improved engagement and followership metrics through feature engineering, value-tree tuning, and online A/B testing of neural recommendation models.

Verneek AI Jun 2023 – Jan 2024

Machine Learning Researcher/Engineer, Personalized Search *New York, NY*

Built a personalized multimodal search system and its evaluation framework, supporting personalization through textual preferences and user interactions; fine-tuned embedding models on synthetic e-commerce interaction data.

Walmart Aug 2018 – Jul 2021

Applied Scientist 2, Voice Assistant for E-Commerce *Bangalore, India*

Indic Speech Recognition. Improved ASR with hierarchical CTC models and joint fusion of neural LMs with Transducer loss, giving a ∼38% relative WER reduction; improved decoding speed using algorithmic approximations.

Indic Speech Synthesis. Built a multilingual TTS framework with automatic transliteration and learnable sentence-style embeddings for conversational prosody, beating commercial APIs on code-mixed and vernacular text.

Microsoft May 2017 – Aug 2017

Software Engineering Intern, Microsoft Excel *Hyderabad, India*

Developed UI and backend functionality for PivotTables in Excel for macOS; received a full-time return offer.

Selected Open Source

Repository titles are linked.

RNN-Transducer Prefix Beam Search: Optimized prefix-beam search for speech recognition model with caching, batching, and 2-D beam pruning. >10x speed-up. Studied the increment in error-rates caused by these approximations. Open-sourced CUDA implementation (67 stars).

RNN-Transducer Loss Function: Devised diagonal parallelism to reduce time complexity from $\mathcal{O}(T * U)$ to $\mathcal{O}(T + U)$. Open-sourced TensorFlow implementation as a Python package (46 stars).

Teaching Experience

CS 441 Applied Machine Learning | *University of Illinois Urbana-Champaign* Fall 2026

CS 546 Advanced Topics in Natural Language Processing | *University of Illinois Urbana-Champaign* Fall 2025

CS 7643 Deep Learning | *Georgia Institute of Technology* Spring 2023

CSE 6740 Computational Data Analysis | *Georgia Institute of Technology* Fall 2022

CS 7650 Natural Language Processing | *Georgia Institute of Technology* Spring 2022

CS 4641 Machine Learning | *Georgia Institute of Technology* Fall 2021

CSO 101 Computer Programming and Linux | *IIT Varanasi* Fall and Spring 2017

Technical Skills

Languages: `Python`, `Bash`, `C++`, `C`, `Java`.

Technologies: `PyTorch`, `JAX`, `Hugging Face`, `vLLM`, `Axolotl`, `TRL`, `VeRL`, `CUDA`, `DeepSpeed`, `Megatron-LM`, `TGI`, `NVIDIA Triton`, `TensorBoard`, `Numpy`, `Pandas`, `Scikit-Learn`, `Matplotlib`, `TensorFlow`, `Docker`, `Kubernetes`, `Helm`, `FastAPI`, `gRPC`, `Django`, `Flask`, `AWS`, `GCP`, `Terraform`, `Git`, LaTeX.

Selected Honors

Outstanding Paper Award, KnowFM Workshop at ACL 2026.

Best Paper Honorable Mention, ACL 2023.

ETH Zürich Summer Research Fellowship (<0.8% acceptance), 2022.

Best Team Award, Flipkart (Walmart), for production Text-to-Speech in Indic languages, 2021.

Winner, FinSBD-2 Shared Task at FinNLP@IJCAI, 2020.

Candidate Master on Codeforces (max rating 1920); ACM ICPC India Regionalist, 2016–17.